



Accuracy Test # 21

Based on NTA Abhyas (PRE-MEDICAL)

**UNITS, DIMENSIONS &
ERRORS**

1. The heat generated in a circuit is given by $Q = I^2 R t$, where I is current, R is resistance and t is time. If the percentage errors in measuring I , R , and t are 2%, 1% and 1% respectively, then the maximum error in measuring heat will be

- (1) 2%
- (2) 3%
- (3) 4%
- (4) 6%

2. The square root of the product of inductance and capacitance has the dimension of

- (1) Length
- (2) Mass
- (3) Time
- (4) Frequency

3. One division on the main scale of vernier callipers is equal to 1 mm. Its vernier scale has 20 equal divisions which match with 16 main scale division. The least count of the vernier calliper is-

- (1) 0.02 mm
- (2) 0.2 mm
- (3) 0.8 mm
- (4) 0.08 mm

4. Dimensional formula of $\frac{a}{b}$ satisfying $P = \frac{a-t^2}{bx}$ where P is the pressure, x is the distance and t is time will be

- (1) $M L^1 T^{-1}$
- (2) $M^2 L T^{-1}$
- (3) $M L^0 T^{-2}$
- (4) $M^0 L^{-1}$

5. The random error in the arithmetic mean of 100 observations is x . Then the random error in the arithmetic mean of 400 observations would be –

- (1) $4x$
- (2) $x/4$
- (3) $2x$
- (4) $x/2$

6. A, B, C and D are four different physical quantities having different dimensions. None of them is dimensionless. But we know that the equation $AD = C \ln(BD)$ holds true. Then which of the combination is not a meaningful quantity?

- (1) $\frac{C}{BD} - \frac{A^2 D^2}{C}$
- (2) $A^2 - B^2 C^2$
- (3) $\frac{A}{B} - C$
- (4) $\frac{A^2 - AC}{D}$

7. Which of the following physical quantities has neither dimensions nor unit?

- (1) Angle
- (2) Luminous intensity
- (3) Coefficient of friction
- (4) Current

8. The mean time period of second's pendulum is 2.00 s and mean absolute error in the time period is 0.05 s. To express maximum estimate of error, the time period should be written as

- (1) $(2.00 \pm 0.01) s$
- (2) $(2.00 \pm 0.025) s$
- (3) $(2.00 \pm 0.05) s$
- (4) $(2.00 \pm 0.10) s$

9. If the diameter of a cylinder is $(12.6 \pm 0.1) \text{ cm}$ and its height is $(34.2 \pm 0.1) \text{ cm}$, then find the volume of the cylinder to the nearest significant figure.

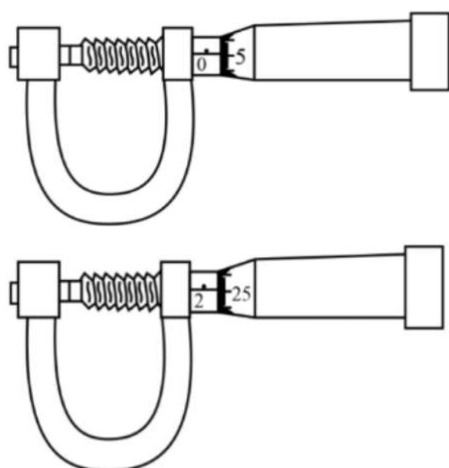
- (1) $4264.4 \pm 81 \text{ cm}^3$
- (2) $4260 \pm 81 \text{ cm}^3$



(3) $4300 \pm 80 \text{ cm}^3$

(4) $4260 \pm 80 \text{ cm}^3$

- 10.** The figure shows two diagrams of the same screw gauge. In the first case, nothing is kept in its jaw and in the second case a small ball is kept between the jaws for which the diameter is required to be measured. The number of circular divisions on the shown screw gauge is 50. It moves 0.5mm on the main scale for one complete rotation and the main scale has $\frac{1}{2}$ mm marks. The diameter of the ball is-



- (1) 2.25 mm
 (2) 2.20 mm
 (3) 1.20 mm
 (4) 1.25 mm

- 11.** If the error in the measurement of momentum is 20 %, then the error in the calculation of kinetic energy is (Assume the error in measurement of mass as zero)-

- (1) 20%
 (2) 44%
 (3) 40%
 (4) 200%

- 12.** In the question

$$\left(p + \frac{a}{V^2}\right)(V - b) = \text{constant}$$

The unit of a is -

- (1) $\text{dyne} \times \text{cm}^5$
 (2) $\text{dyne} \times \text{cm}^4$
 (3) $\text{dyne} \times \text{cm}^{-3}$
 (4) $\text{dyne} \times \text{cm}^{-2}$

- 13.** In a new system of units, the unit of force is 100N, unit of length is 10 m and unit of time is 100s. Unit of mass in this system can be expressed as-

- (1) 10^5 Kg
 (2) 10^6 kg
 (3) 10^2 kg
 (4) 10^3 kg

- 14.** The least count of a stopwatch is 0.2 s. The time of 20 oscillations of a pendulum is measured to be 25 s. The percentage error in the measurement of the time period will be-

- (1) 8 %
 (2) 1.8 %
 (3) 0.8 %
 (4) 0.1 %

- 15.** Select the dimensional formula of $\frac{B^2}{2\mu_0}$ -

- (1) $[M^1 L^1 T^2]$
 (2) $[M^{-1} L^1 T^2]$
 (3) $[M^{-1} L^{-1} T^{-2}]$
 (4) $[M^1 L^{-1} T^{-2}]$

- 16.** If the velocity of light (c), gravitational constant (G) and Planck's constant (h) are chosen as fundamental units, then the dimensions of mass in this new system is-

- (1) $c^{\frac{1}{2}} G^{\frac{1}{2}} h^{\frac{1}{2}}$
 (2) $c^{1/2} G^{1/2} h^{-1/2}$
 (3) $c^{1/2} G^{-1/2} h^{1/2}$
 (4) $c^{-1/2} G^{1/2} h^{1/2}$

17. The pressure on a circular plate is measured by measuring the force on the plate and radius of the plate. If the errors in measurement of force and radius are 5% and 3% respectively, then the percentage of error in the measurement of pressure is-

- (1) 8
- (2) 14
- (3) 11
- (4) 12

18. The relative error in the determination of the surface area of a sphere is α . The relative error in the determination of its volume is-

- (1) $\frac{3}{2}\alpha$
- (2) $\frac{2}{3}\alpha$
- (3) α
- (4) $\frac{5}{2}\alpha$

19. The dimensional formula of $\frac{1}{\mu_0 \epsilon_0}$ is –

- (1) $[M^0 L^2 T^{-2}]$
- (2) $[M^0 L^1 T^{-1}]$
- (3) $[M^0 L^{-2} T^{-2}]$
- (4) $[M^0 L^1 T^{-2}]$

20. The unit of energy if fundamental units are taken as force (F), acceleration (A) and time (T) is-

- (1) $[F A T^2]$
- (2) $[FA^2 T]$
- (3) $[FA^2 T^2]$
- (4) $[FAT]$

21. The voltage and current in a conductor are measured $(50 \pm 2)V$ and $(5 \pm 0.2)A$. The percentage error in the calculation of resistance is-

- (1) 4%
- (2) 5%

- (3) 7%
- (4) 8%

22. For a screw gauge least count of the main scale is 1 mm. Find the minimum number of divisions on the circular scale so that it can measure a diameter of $5\mu m$ -

- (1) 500
- (2) 100
- (3) 50
- (4) 200

23. The frequency f of vibration of mass m suspended from a spring of spring constant k is given by $f = cm^x k^y$. Where c is a dimensionless constant. The values of x and y are respectively

- (1) $\frac{1}{2}, -\frac{1}{2}$
- (2) $-\frac{1}{2}, \frac{1}{2}$
- (3) $\frac{1}{2}, \frac{1}{2}$
- (4) $-\frac{1}{2}, -\frac{1}{2}$

24. Energy (E), angular momentum (L) and universal gravitational constant (G) are chosen as fundamental quantities. The dimensions of universal gravitational constant in the dimensional formula of Planck's constant (h) is-

- (1) zero
- (2) -1
- (3) $5/3$
- (4) 1

25. The distance moved by the screw of a screw gauge is 2mm in four rotations and there are 50 divisions on its cap. When nothing is put between its jaws, 30th division of circular scale coincides with the reference line, with zero of circular scale lying above the reference line. When a plate is placed between the jaws, the main scale reads 2 divisions and the circular scale reads 20 divisions. The thickness of the plate is-

- (1) 1.5 mm
 (2) 1.2 mm
 (3) 1.4 mm
 (4) 1.6 mm
- 26.** There is an unknown quantity, $x = R \left(\frac{l}{100} \right)$ here l is the length (in cm) measured using a scale having least count of 1 cm. and R is a quantity known accurately. Find the percentage error in measurement of x for $l = 50$ cm.
- (1) 1 %
 (2) 6 %
 (3) 2 %
 (4) 4 %
- 27.** What is the dimensions of the coefficient of viscosity η ?
- (1) $ML^{-1}T^{-2}$
 (2) MLT^{-1}
 (3) $ML^{-1}T^{-1}$
 (4) $ML^{-2}T^{-2}$
- 28.** The respective number of significant figures for the numbers 23.023, 0.0003 and 2.1×10^{-3} are: -
- (1) 4, 4, 2
 (2) 5, 1, 2
 (3) 5, 1, 5
 (4) 5, 5, 2
- 29.** An athletic coach told his team that muscle multiplied with speed equal to power. What dimensions does he view for muscle?
- (1) MLT^{-2}
 (2) $ML^2 T^{-2}$
 (3) MLT^{-3}
 (4) L
- 30.** The external and internal diameters of a hollow cylinder are measured with the help of vernier callipers. Their values are 4.23 ± 0.01 cm and 3.87 ± 0.01 cm respectively. The thickness of the wall of the cylinder is -
- (1) 0.36 ± 0.02 cm
 (2) 0.18 ± 0.02 cm
 (3) 0.36 ± 0.01 cm
 (4) 0.18 ± 0.01 cm
- 31.** A vernier caliper has 20 divisions on the vernier scale which coincide with 19 divisions on the main scale. The least count of the instrument is 0.1 mm. The main scale divisions are of
- (1) 0.5 mm
 (2) 1 mm
 (3) 2 mm
 (4) 1/4 mm
- 32.** The time period of oscillation of a simple pendulum is given by $T = 2\pi \sqrt{\frac{l}{g}}$. The length of the pendulum is measured as $l = 10 \pm 0.01$ cm and the time period as $T = 0.5 \pm 0.02$ s. The percentage error in the value of g is
- (1) 5%
 (2) 8%
 (3) 7%
 (4) none of these
- 33.** A public park, in the form of a square, has an area of $(100 \pm 0.2) m^2$. The side of park is
- (1) $(10 \pm 0.01) m$
 (2) $(10 \pm 0.1) m$
 (3) $(10.0 \pm 0.1) m$
 (4) $(10.0 \pm 0.2) m$
- 34.** If E , m , J and G represent energy, mass, angular momentum and gravitational constant respectively, then the dimensional formula $\frac{EJ^2}{m^5 G^2}$ is

(1) $[MLT^{-2}]$

(2) $[M^0L^0T]$

(3) $[M^0L^2T^0]$

(4) Dimensionless

35. The set of physical quantities among the following which is dimensionally different is:

(1) Terminal velocity, drift velocity, critical velocity

(2) Potential energy, work done, kinetic energy

(3) Disintegration constant, frequency, angular velocity

(4) Dipole moment, electric flux, electric field

36. If C' be the capacitance and V be the electric potential, then the dimensional formula of CV^2 is

(1) $[ML^{-3}TA]$

(2) $[M^0LT^{-2}A^0]$

(3) $[ML^1T^{-2}A^{-1}]$

(4) $[ML^2T^{-2}A^0]$

37. The dimensional formula for permittivity of free space ϵ_0 in the equation

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1q_2}{r^2}$$

Where symbols have their usual meaning, is

(1) $[M^1L^3A^{-2}T^{-4}]$

(2) $[M^{-1}L^{-3}A^2T^4]$

(3) $[M^1L^{-3}A^{-2}T^{-4}]$

(4) $[M^1L^3A^{-4}T^2]$

38. Unit of magnetic moment is

(1) Ampere — metre²

(2) Ampere — metre

(3) Weber — metre²

(4) Weber — metre

39. The density of a liquid in the CGS system is 0.625 g cm^{-3} . What is its magnitude in SI system?

(1) 0.625

(2) 0.0625

(3) 0.00625

(4) 625

40. The quantities A and B are related by the relation, $m = \frac{A}{B}$, where m is the linear density and A is the force. The dimensions of B are of

(1) Pressure

(2) Work

(3) Latent heat

(4) None of the above

41. When the length of the vibrating segment of a Sonometer wire is increased by 1% the percentage changes its frequency is- (Given - The frequency of Sonometer wire is

$$v = \frac{1}{2l} \sqrt{\frac{T}{m}}$$

(1) $\frac{100}{101}$

(2) $\frac{99}{100}$

(3) 1

(4) 2

42. A force is given by $F = at + bt^2$ where t is time. The dimensions of a and b are-

(1) $[MLT^{-4}]$, $[MLT^{-2}]$

(2) $[MLT^{-3}]$, $[MLT^{-4}]$

(3) $[ML^2T^{-3}]$, $[ML^2T^{-2}]$

(4) $[ML^2T^{-3}]$, $[ML^2T^{-4}]$

43. Among the following quantities, the quantity whose dimension is independent of mass and length is-

(1) Energy

(2) Thrust

- (3) Momentum
- (4) Angular velocity

44. A screw gauge gives the following reading when used to measure the diameter of a wire. Main scale reading: 0 mm. Circular scale reading: 52 divisions Given that 1 mm on main scale corresponds to 100 divisions of the circular scale. The diameter of wire from the above data is: -

- (1) 0.026 cm
- (2) 0.005 cm
- (3) 0.52 cm
- (4) 0.052 cm

45. When a copper sphere is heated, maximum percentage change will be observed in—

- (1) Radius
- (2) Area
- (3) Volume
- (4) None of these